Homework 5 (due 4/8/2026)

1. For 3D carriers with parabolic energy bands, please work out the 3D density of states. (10ps)
2. For 2D carriers with parabolic energy bands, please work out the 2D density of states. (10ps)
3. For 1D carriers with parabolic energy bands, please work out the 1D density of states. (10ps)
4. For the 3D carriers with parabolic energy bands, the density of states goes as $\sqrt{E}$. Assume a non-parabolic band described by the so-called Kane dispersion,

$$E(k)\left[1+\alpha E(k)\right]=\frac{\hbar^2k^2}{2m^*(0)}$$

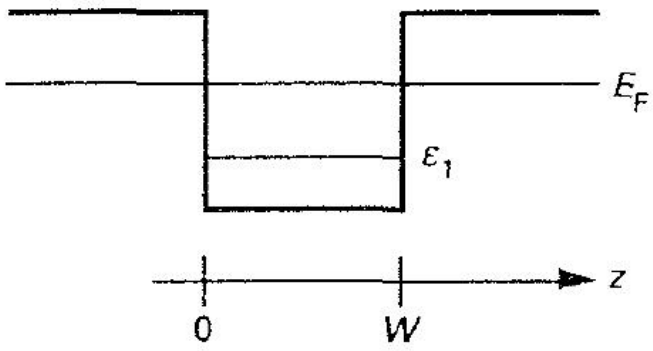
Please work out the corresponding 3D density of states. (10ps)

5. (a) Five electrons exist in a three-dimensional infinite potential well with all three widths equal to a=12 Å. Determine the Fermi energy level at T =0 K. (5ps) (b) Repeat part (a) for 13 electrons. (5ps)
6. Please derive the following statistics of donors and acceptor as shown in lecture note. (20pts)

$$N_D^{empty} \equiv N_D^+ = N_D f_{00} = N_D \frac{1}{1+2e^{(E_F-E_D)/k_BT}}$$

$$N_A^{filled} \equiv N_A^- = N_A f_{0000} = N_A \frac{1}{1+4e^{(E_A-E_F)/k_BT}}$$

7. The following problem concerns electrons in a quantum well of width W, with only one subband at E = ε_1. Assume equilibrium conditions and that the Fermi level E_f is located above ε_1. Answer the following questions assuming parabolic energy bands.



   a) Write an expression, involving sums over momentum space, which gives the average kinetic energy (due to its motion in the x-y plan) per electron. (5ps)
   b) Convert the sum to an integral over momentum or k-space(5ps)
   c) Write an expression, involving an integral over energy space, which give the average kinetic energy per electron. (5ps)
   d) Assume T=0K, and evaluate the average kinetic energy per electron. You may work in either energy or momentum space. (5ps)
8. At room temperature, an intrinsic GaAs has L minima along <111> directions, and the following data are available: Electron effective mass of density state for Γ valley and X valley

are 0.063 m_0 and 0.85m_0. The hole effective mass of density state is 0.53 m_0. **The density of state can be calculated with effective mass of density state using the same formula developed in the class.** At what temperature the number of electrons in the Γ minima and X minima will be equal if the energy separation between them is 0.29 eV? The band gap for GaAs is 1.42 eV. (10ps)

1. $E = \frac{\hbar^2 k^2}{2m^*}$. $k^2 = k_x^2 + k_y^2 + k_z^2$

wave function : $\varphi(x,y,z) \sim e^{ik_x x} \cdot e^{ik_y y} \cdot e^{ik_z z}$

Assume $\varphi(x,y,z) = \varphi(x+a,y,z) + \varphi(x,y+b,z) + \varphi(x,y,z+c)$

$k_x = \frac{2\pi}{a} n_x$. $k_y = \frac{2\pi}{b} \cdot n_y$. $k_z = \frac{2\pi}{c} \cdot n_z$

Consider a cubic space of $a \times b \times c$.

$\Delta V_k = \Delta k_x \cdot \Delta k_y \cdot \Delta k_z = \frac{(2\pi)^3}{abc}$

$D(E) \cdot dE = \frac{4\pi k^2 \cdot dk}{\Delta V_k} \times 2 = \frac{abc\, k^2 dk}{\pi^2}$

$\Rightarrow$ Density of state : $D(E) = abc \cdot \frac{k^2}{\pi^2} \cdot \frac{dk}{dE} = abc \cdot \frac{m^*}{\pi^2 \hbar^3} \sqrt{2m^* E}$

$\Rightarrow$ Reduced Density of state : $g(E) = \frac{D(E)}{V} = \frac{m^* \sqrt{2m^* E}}{\pi^2 \hbar^3}$

2. $E = \frac{\hbar^2 k^2}{2m^*}$. $k^2 = k_x^2 + k_y^2$

wave function $\varphi(x,y) \sim e^{ik_x x} \cdot e^{ik_y y}$. Assume $\varphi(x,y) = \varphi(x+a,y) + \varphi(x,y+b)$

$k_x = \frac{2\pi}{a} \cdot n_x$. $k_y = \frac{2\pi}{b} \cdot n_y$

consider a rectangle space of $a \times b$.

$\Delta S_k = \Delta k_x \cdot \Delta k_y = \frac{(2\pi)^2}{ab}$

$D(E) \cdot dE = \frac{2\pi k \cdot dk}{\Delta S_k} \times 2 = ab \cdot \frac{k}{\pi} dk$

$\Rightarrow$ Density of state : $D(E) = \frac{ab}{\pi} \cdot k \frac{dk}{dE} = ab \cdot \frac{m^*}{\pi \hbar^2}$

$\Rightarrow$ Reduced Density of state : $g(E) = \frac{D(E)}{V} = \frac{m^*}{\pi \hbar^2}$

3. $E = \frac{\hbar^2 k^2}{2m^*}$

$\varphi(x) \sim e^{ikx}$. Assume $\varphi(x) = \varphi(x+a)$

$k = \frac{2\pi}{a} \cdot n$. $\Delta l = \frac{2\pi}{a}$

$D(E) \cdot dE = \frac{dk}{\Delta l} \times 2$

$\Rightarrow$ Density of state : $D(E) = \frac{a}{\pi} \frac{dk}{dE} = \frac{a}{\pi \hbar} \sqrt{\frac{m^*}{2E}}$

$\Rightarrow$ Reduced Density of state : $g(E) = \frac{1}{\pi \hbar} \cdot \sqrt{\frac{m^*}{2E}}$

4. $E(k)[1+\alpha E(k)] = \frac{\hbar^2 k^2}{2m^*(\omega)}$

$$\Rightarrow \frac{\partial k}{\partial E} = \frac{(1+2\alpha E)}{\hbar}\sqrt{\frac{m^*(\omega)}{2E(1+\alpha E)}}, \quad k = \frac{\sqrt{2m^*(\omega)\cdot E(1+\alpha E)}}{\hbar}$$

$$k^2 = k_x^2 + k_y^2 + k_z^2$$

wave function : $\varphi(x,y,z) \sim e^{ik_x x}\cdot e^{ik_y y}\cdot e^{ik_z z}$

Assume $\varphi(x,y,z) = \varphi(x+a,y,z) + \varphi(x,y+b,z) + \varphi(x,y,z+c)$

$$k_x = \frac{2\pi}{a}n_x \quad . \quad k_y = \frac{2\pi}{b}\cdot n_y \quad . \quad k_z = \frac{2\pi}{c}\cdot n_z$$

Consider a cubic space of $a \times b \times c$.

$$\Delta V_k = \Delta k_x \cdot \Delta k_y \cdot \Delta k_z = \frac{(2\pi)^3}{abc}$$

$$D(E)\cdot dE = \frac{4\pi k^2 \cdot dk}{\Delta V_k} \times 2$$

$$\Rightarrow \text{Density of state} : D(E) = abc\cdot\frac{k^2}{\pi^2}\cdot\frac{dk}{dE} = abc\cdot\frac{m^*(\omega)\sqrt{2m^*(\omega)}\cdot\sqrt{E(1+\alpha E)}\cdot(1+2\alpha E)}{\hbar^3\pi^2}$$

$$\Rightarrow \text{Reduced Density of state} : g(E) = \frac{D(E)}{V} = \frac{m^*(\omega)\sqrt{2m^*(\omega)}\cdot\sqrt{E(1+\alpha E)}\cdot(1+2\alpha E)}{\hbar^3\pi^2}$$

5. (a) $V(x,y,z) = \begin{cases} 0, & 0<x<a,\ 0<y<a,\ 0<z<a \\ \infty, & \text{otherwise} \end{cases}$

while $0<x<a,\ 0<y<a,\ 0<z<a$, $-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}\right) = E\,\psi(x)\cdot\psi(y)\cdot\psi(z)$

$\Rightarrow E_x = \frac{\hbar^2 k_x}{2m}$, $E_y = \frac{\hbar^2 k_y}{2m}$, $E_z = \frac{\hbar^2 k_z}{2m}$, $k_x = \frac{n_x\pi}{a}$, $k_y = \frac{n_y\pi}{a}$, $k_z = \frac{n_z\pi}{a}$

$\Rightarrow E_{n_x n_y n_z} = \frac{\hbar^2\pi^2}{2ma^2}(n_x^2+n_y^2+n_z^2)$

At $T = 0\,k$

The lowest energy state : $E_{111} = \frac{3\hbar^2\pi^2}{2ma^2}$, which can place 2 electrons.

The rest 3 electrons can be placed at E_{112}, E_{121}, E_{211} $= \frac{3\hbar^2\pi^2}{ma^2}$

$\Rightarrow$ Fermi energy level $E_{Fi} \approx 1.57\,eV$

(b) for 13 electrons.

2 electrons will occupy E_{111}. 6 electrons will occupy E_{112}, E_{121}, E_{211}.

The rest 5 electrons will be placed at E_{122}, E_{212} or E_{221}. $= \frac{9\hbar^2\pi^2}{2ma^2}$

$\Rightarrow$ Fermi energy level $E_{Fi} \approx 2.35\,eV$

6. For Doner. Assume the Doping energy level is E_D, number of Doner is N_D, N_D^+ doners donate a electron. we have $N = N_D - N_D^+$ electrons left.

Number of states : $W = \frac{N_D!}{N_A^+! \cdot N!} \cdot g^N$ for Doner, $g = 2$

And we have $\sum_i N_i = N$, $\sum_i N_i E_i = N \cdot E_D$

Applying Lagrange multiplier

$\mathcal{L} = \ln W - \alpha N - \beta \cdot N \cdot E_D$

$\mathcal{L} = \ln N_D! - \ln N_A^+! - \ln N! + \ln 2^N - \alpha N - \beta \cdot N \cdot E_D$

$\frac{\partial \mathcal{L}}{\partial N} = 0 \Rightarrow \frac{N_D - N}{N} = \frac{1}{1 + \frac{1}{2}\exp(\alpha + \beta E_D)}$

From thermodynamics, we know $\alpha = -\frac{E_F}{kT}$, $\beta = \frac{1}{kT}$

$\Rightarrow \frac{N_D - N}{N} = \frac{N_D^+}{N_D - N_D^+} = \frac{1}{1 + \frac{1}{2}\exp(\frac{E_D - E_F}{kT})}$

$\Rightarrow \frac{N_D^+}{N_D} = \frac{1}{1 + 2\exp(E_F - E_D)/k_B T}$

$N_D^{empty} \equiv N_D^+ = N_D \cdot f_{oo} = N_D \frac{1}{1 + 2\exp(E_F - E_D)/k_B T}$

For Acceptor, Assume the Doping energy level is E_A, number of Acceptor is N_A, N_A^- Acceptors accepts a electron. we have $N = N_A - N_A^-$ holes left.

take $g = 4$, similarly, we have $\frac{N_A^-}{N_A} = \frac{1}{1 + 4\exp(\frac{E_A - E_F}{k_B T})}$

$N_A^{filled} \equiv N_A^- = N_A \cdot f_{oooo} = N_A \frac{1}{1 + 4\exp(\frac{E_A - E_F}{k_B T})}$

7. a) In x-y plane. Kinetic energy for electron: $E_k = \frac{\hbar^2}{2m^*}(k_x^2 + k_y^2)$

Total energy: $E_{total} = E_k + \varepsilon_1 = \frac{\hbar^2(k_x^2 + k_y^2)}{2m^*} + \varepsilon_1$

$$f(E) = \frac{1}{1 + \exp[(E - E_F)/k_B T]}$$

$$\langle E_k \rangle = \frac{\sum_{k_x=1}^{+\infty}\sum_{k_y=1}^{+\infty} E_k \cdot f(E_{total})}{\sum_{k_x=1}^{+\infty}\sum_{k_y=1}^{+\infty} f(E_{total})}$$

$$\Rightarrow \langle E_k \rangle = \frac{\sum_{k_x=1}^{+\infty}\sum_{k_y=1}^{+\infty}\left[\frac{\hbar^2(k_x^2 + k_y^2)}{2m^*} \cdot \frac{1}{1 + \exp[(\frac{\hbar^2(k_x^2+k_y^2)}{2m^*} + \varepsilon_1 - E_F)/k_B T]}\right]}{\sum_{k_x=1}^{+\infty}\sum_{k_y=1}^{+\infty}\frac{1}{1 + \exp[(\frac{\hbar^2(k_x^2+k_y^2)}{2m^*} + \varepsilon_1 - E_F)/k_B T]}}$$

(b) $$\langle E_k \rangle = \frac{\int_0^{+\infty}\int_0^{+\infty} E_k \cdot f(E_{total}) \cdot dk_x\, dk_y}{\int_0^{+\infty}\int_0^{+\infty} f(E_{total})\, dk_x\, dk_y}$$

$$\Rightarrow \langle E_k \rangle = \frac{\int_0^{+\infty}\int_0^{+\infty} \frac{\hbar^2(k_x^2 + k_y^2)}{2m^*} \cdot \frac{1}{1 + \exp[(\frac{\hbar^2(k_x^2+k_y^2)}{2m^*} + \varepsilon_1 - E_F)/k_B T]}\, dk_x\, dk_y}{\int_0^{+\infty}\int_0^{+\infty} \frac{1}{1 + \exp[(\frac{\hbar^2(k_x^2+k_y^2)}{2m^*} + \varepsilon_1 - E_F)/k_B T]}\, dk_x\, dk_y}$$

(c) $$\langle E_k \rangle = \frac{\int_{\varepsilon_1}^{+\infty}(E - \varepsilon_1) \cdot f(E)\, dE}{\int_{\varepsilon_1}^{+\infty} f(E)\, dE}$$

$$\Rightarrow \langle E_k \rangle = \frac{\int_{\varepsilon_1}^{+\infty}(E - \varepsilon_1) \cdot \frac{1}{1 + \exp[(E - E_F)/k_B T]}\, dE}{\int_{\varepsilon_1}^{+\infty}\frac{1}{1 + \exp[(E - E_F)/k_B T]}\, dE}$$

(d) While $T = 0\,k$. $f(E) = \begin{cases} 1, & E < E_F \\ 0, & E > E_F \end{cases}$

$$\langle E_k \rangle = \frac{\int_{\varepsilon_1}^{+\infty}(E - \varepsilon_1) \cdot f(E)\, dE}{\int_{\varepsilon_1}^{+\infty} f(E)\, dE} = \frac{\int_{\varepsilon_1}^{E_F}(E - \varepsilon_1)\, dE}{\int_{\varepsilon_1}^{E_F} dE}$$

$$\Rightarrow \langle E_k \rangle = \frac{E_F - \varepsilon_1}{2}$$

8. While $E_c - E_F > 3k_BT$, $f(E) \cong \exp\left(-\frac{E-E_F}{k_BT}\right)$

At the bottom of the conduction band: $g_c(E) = \frac{m_n^* \cdot \sqrt{2m_n^*} \cdot \sqrt{E-E_c}}{\pi^2 \hbar^3}$

$$n = \int_{E_c}^{+\infty} g_c(E) \cdot f(E) \cdot dE = \int_{E_c}^{+\infty} \exp\left(-\frac{E-E_F}{k_BT}\right) \frac{m_n^* \cdot \sqrt{2m_n^*} \cdot \sqrt{E-E_c}}{\pi^2 \hbar^3} dE$$

$$\Rightarrow n = 2\left(\frac{2\pi m_n^* k_B T}{\hbar^2}\right)^{\frac{3}{2}} \exp\left(-\frac{E_c - E_F}{k_BT}\right)$$

Since $3n_T = n_X$.

$$\Rightarrow 3 \times 2\left(\frac{2\pi m_{n,T}^* k_B T}{\hbar^2}\right)^{\frac{3}{2}} \exp\left(-\frac{E_{c,T}-E_F}{k_BT}\right) = 2\left(\frac{2\pi m_{n,X}^* k_B T}{\hbar^2}\right)^{\frac{3}{2}} \exp\left(-\frac{E_{c,X}-E_F}{k_BT}\right)$$

$$\Rightarrow 3\left(\frac{m_{n,T}^*}{m_{n,X}^*}\right)^{\frac{3}{2}} = \exp\left(-\frac{E_{c,X}-E_{c,T}}{k_BT}\right)$$

$m_{n,T}^* = 0.063\, m_0$, $m_{n,X}^* = 0.85\, m_0$, $E_{c,X} - E_{c,T} = 0.29\,eV$

$$\Rightarrow T = -\frac{E_{c,X}-E_{c,T}}{k_B \cdot \ln 3 \cdot \left(\frac{m_{n,T}^*}{m_{n,X}^*}\right)^{\frac{3}{2}}}$$

$$T \approx \sout{862\,K}$$